

Realizing scalar comparison profiles by finite uniform dice

Research proof; independent mathematical review completed

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No bound on the number of faces is asserted in this note. The point is to transfer robust local comparison profiles from a scalar quadrature measure to a completely finite, exactly permutation-fair family.

Theorem 1. *Let $m \geq 2$ and let σ be a probability measure on $[0, 1]$ whose moments through degree m equal those of Lebesgue measure. Fix $t \in (0, 1)$ with $\sigma(\{t\}) = 0$, and let $\rho > 0$. There is a permutation-fair family of $m + 1$ finite uniform dice, with pairwise distinct face labels, and a face of one die D at which the probabilities p_i that the other dice fall below that face satisfy*

$$|p_1 - F_\sigma(t)| < \rho, \quad |p_i - t| < \rho \quad (2 \leq i \leq m).$$

In particular the weights of all rows of the comparison tensor of D are equal. The remaining m dice can be partitioned into one special die and $m - 1$ dice with the same number of faces as D .

Lemma 1. *Under the hypotheses of Theorem 1, for every $\eta > 0$ there is a degree- m equal-weight quadrature for Lebesgue measure, with $K \geq m^2$ distinct nodes in $(0, 1)$, none equal to t , and empirical measure ν satisfying $|F_\nu(t) - F_\sigma(t)| < \eta$.*

Proof. We first approximate σ by a bounded, strictly positive density with exactly the same first m moments. Smooth each point of σ on a small interval contained in $(0, 1)$, including points near the endpoints, to obtain a bounded density g_a converging weakly to σ as $a \downarrow 0$. For example, push the measure into $[a, 1 - a]$ and convolve with a uniform kernel of radius $a/2$. Fix $\delta > 0$ and form $(1 - \delta)g_a + \delta$. Its moment error against any fixed polynomial tends to zero as $a \downarrow 0$.

Let $\phi_0 = 1, \phi_1, \dots, \phi_m$ be the orthonormal shifted Legendre polynomials. Subtract the polynomial

$$h_a = \sum_{j=1}^m \phi_j \int_0^1 \phi_j(x)((1 - \delta)g_a(x) + \delta) dx.$$

For fixed m, δ , its supremum norm tends to zero. Choose a so small that $\|h_a\|_\infty < \delta/2$. Then

$$w = (1 - \delta)g_a + \delta - h_a$$

is a bounded density, bounded below by $\delta/2$, and has the required exact moments. By first making δ small and then a small, $w(x) dx$ can be made arbitrarily close to σ weakly.

A density bounded above and below by positive constants is a doubling weight. The equal-weight quadrature theorem of Gilboa–Peled (Theorem 1.1) supplies quadratures ν_D for w of arbitrarily high degree D , with arbitrarily large admissible cardinality. As $D \rightarrow \infty$, these empirical measures converge weakly to $w(x) dx$: approximate a continuous test function uniformly by a polynomial, and use exactness. For $D \geq m$, each ν_D is also exact through degree m for Lebesgue measure.

Choose $D \geq 2m + 2$ and cardinality $K \geq m^2$. Such a rule has at least $m + 2$ distinct support points; otherwise the square of its support-vanishing polynomial contradicts exactness and positivity of w . At least m support points are interior. Select m such nodes as pivot coordinates. The first m moment equations have an invertible Vandermonde Jacobian in the pivots. The implicit function theorem permits arbitrarily small perturbations of all other coordinates, preserving the first m moments by compensating with the pivots. Endpoints can be moved inward, and all collisions can be avoided, as in the regularization lemma in `size_bounds/polynomial_exceptional_die.tex`.

We may also avoid the fixed value t . For a pivot initially equal to t , vary a free coordinate at a support point different from all pivots; there is such a point because the original rule has at least $m + 2$ distinct support points. The corresponding pivot derivative is minus a nonzero Lagrange cardinal value. Thus the constraint that this pivot equal t is a proper analytic zero set. The same is immediate for free coordinates. A finite union of these zero sets and the collision sets cannot cover the permitted open set of inward perturbations.

These regularizations can be arbitrarily small, so they preserve weak approximation. Choose the preceding parameters successively so that the resulting empirical measure is close enough to σ at the continuity point t . More explicitly, first choose two continuity points $t_- < t < t_+$ of σ whose distribution values differ by less than $\eta/4$. Weak approximation makes the distribution values of the quadrature at both endpoints close to those of σ ; a perturbation of size less than half their distance to t brackets the new value at t between these values. This proves the lemma. $\square$

Proof of Theorem 1. Apply Lemma 1 with error smaller than $\rho/4$, and write its nodes as $u_1 < \dots < u_K$. The one-exceptional-die construction in `size_bounds/polynomial_exceptional_die.tex` applies to any such regular rule with $K \geq m^2$, not just to its initial $O(m^2)$ -node choice. Its proof has the following useful additional freedom: for every $\gamma > 0$, the nodes can be replaced by rational v_q arbitrarily close to u_q , and the m old continuous dice can be given positive rational step densities f_j such that

$$\|f_j - 1\|_\infty < \gamma \quad (1 \leq j \leq m),$$

while their joint order law with the uniform special die on the v_q is exactly fair. Indeed, the fixed separated bump functions are bounded, and their rational Vandermonde coefficients tend to zero as $v \rightarrow u$. Choose the rational approximation close enough that no node crosses t .

The final finite realization of that construction partitions the interval into rational constant-density cells and replaces every cell by an equal-sided fair block with rational mass multiplicities. It permits any further rational refinement of the partition. Take a refinement of mesh at most τ , and select a cell J within distance τ of t , contained in the same gap between special nodes as t . Such a positive-length cell exists even if t is already a density breakpoint. All old densities are positive, so every old die, including a selected die D , has faces in J . Place the discrete block's faces at distinct real coordinates inside their cells before replacing the total order by integer labels.

Let x be a face of D in J . Each other old die has the correct mass in every cell. Its finite CDF differs from its continuous CDF at any point by at most the mass of that point's cell, which is at most $(1 + \gamma)\tau$. Its continuous CDF differs from the identity by at most γ . Therefore

$$|p_i - t| \leq \gamma + (2 + \gamma)\tau \quad (2 \leq i \leq m).$$

The special die has no node between x and t , so $p_1 = F_\nu(t)$. Choose γ, τ small enough. This proves all assertions, including exact fairness and uniform face weights. $\square$

Limit of the statement. The approximation parameters, the doubling constant, the quadrature cardinality, and the final denominator clearing may depend arbitrarily badly on m and ρ . This is a realization theorem for robust inequalities, not an efficient construction.